Electronic Appliance Store Management System

Database Management Systems (DBMS) — Review 2: Database Implementation & Backend Integration

Item	Detail
Student	Sanjiv Shine Fernandas
Subject	Database Management Systems
Evaluation	Review 2 — Database Implementation & Backend Integration
Database	Oracle Database 21c XE, pluggable database `XEPDB1`, schema `appliance_app`
Backend	Python Flask, python-oracledb (Thin mode)
Frontend	HTML5, CSS3, vanilla JavaScript
Environment	Windows local demonstration

> **Submission:** repository <https://github.com/sanjiv283/Electronic_Appliance_Store_Manager> (default branch `main`). Companion files: `docs/Review_2_Presentation.pptx`, `docs/demo_guide.md`, `README.md`.

> **Verification disclosure.** The report and code were authored in an environment with no Oracle Database. On 2026-10-22 the student ran the 16 live integration tests (INT-01..16) on a real Oracle Database 21c XE (`XEPDB1`, separate test schema): **16 PASSED**, and executed all 16 queries in `queries.sql` on the demo schema without error. The offline checks in §24 remain as listed. The live results were transcribed from the student's console output into `docs/test_results.xml`; they were not produced by the authoring environment. Anything not covered by those two runs is labelled in §24 and §28.

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1. Abstract

The Electronic Appliance Store Management System is a database-driven web application for a retail appliance shop. A normalized Oracle 21c schema of seven tables stores categories, suppliers, products, customers, employees, sales and sale items. A Flask service layer using python-oracledb provides CRUD for four master entities and transactional multi-item sales with row-level inventory locking. Oracle-side logic consists of one view, one stored procedure, one stored function and one integrity trigger. Sixteen documented SQL queries demonstrate joins, subqueries, aggregation and analytic features. The application connects with a least-privilege account and uses bind variables for all SQL. 76 automated tests ran and passed without Oracle (fake connection), and the 16 live-Oracle integration tests passed on a real Oracle 21c XE instance (run by the student on 2026-10-22).

2. Introduction

Appliance retailers handle high-value items with changing prices and limited stock. Spreadsheet records lead to oversold items, lost price history and inconsistent customer data. A relational database with enforced integrity and transactional sales processing addresses these issues. This project implements that system for Review 2, which evaluates database creation, keys and constraints, sample data, queries, advanced objects, connectivity and CRUD/transactions.

3. Problem statement

Design and implement a relational database and web front end that (a) stores store master data without redundancy, (b) records sales with historical prices, (c) prevents negative or oversold inventory even with simultaneous users, and (d) answers business questions (revenue, best sellers, low stock, customer spend).

4. Objectives

• Create a normalized (3NF) Oracle schema with full integrity constraints.

• Provide deterministic fictional sample data.

• Provide exactly 16 documented queries and four advanced database objects.

• Connect Flask to Oracle securely with python-oracledb Thin mode.

• Implement CRUD and an atomic sales transaction that never oversells.

• Provide automated tests with honest, status-labelled results.

5. Scope and limitations

In scope: master-data CRUD, sales entry and history, restock, backend–Oracle connectivity. Out of scope for Review 2: user authentication/authorization, dashboards and report screens, returns and refunds, taxes/discounts, purchase orders to suppliers, multi-currency, employee CRUD screens (employees are seeded and selectable on sales). Further limitations in §26.

6. Functional requirements

• FR1 Manage products, categories, suppliers, customers (create, list/search, edit, delete).

• FR2 Record a sale with one or more products; totals and prices come from the database.

• FR3 Deduct stock exactly once per sale; reject insufficient stock.

• FR4 Restock a product.

• FR5 Browse sales history and sale details; search by invoice, customer or employee.

• FR6 Health endpoint reporting real database connectivity.

7. Non-functional requirements

• Integrity: all rules enforced in the database where possible (PK/FK/UNIQUE/CHECK).

• Security (database access): no secrets in code, bind variables, least privilege.

• Reliability: transactions commit atomically or roll back; connections always closed.

• Maintainability: service-layer separation; tests per layer.

• Portability: Windows 10/11 local setup.

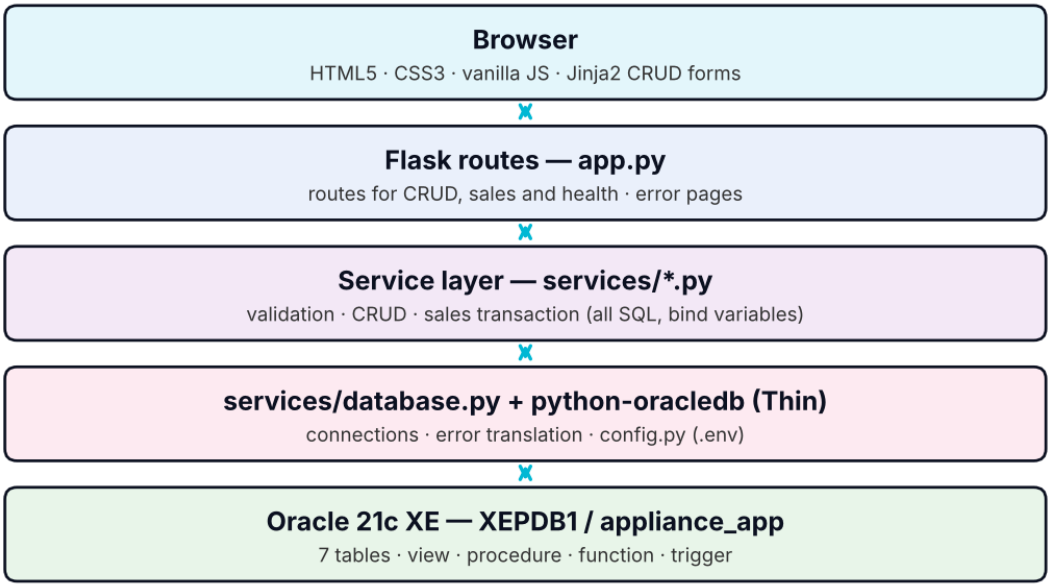
8. Software and hardware requirements

Item	Requirement
OS	Windows 10/11 (development sandbox was Linux, used only for non-Oracle checks)
Python	3.11+ (3.13 used in the sandbox)
Python packages	Flask, oracledb, python-dotenv, pytest (python-docx, reportlab for report build)
Database	Oracle Database 21c XE with `XEPDB1`
Tools	SQL*Plus (bundled with XE), PowerShell, a modern browser
Hardware	2+ CPU cores, 4 GB RAM minimum (Oracle XE limit 2 GB database RAM), ~5 GB disk

Technology stack

Layer	Technology
Presentation	HTML5, CSS3, vanilla JavaScript, Jinja2 templates (CRUD forms)
Application	Python, Flask, `python-dotenv`, service-layer modules
Database access	`python-oracledb` Thin mode (no Oracle Client install needed)
Database	Oracle Database 21c XE, PDB `XEPDB1`, PL/SQL
Quality	pytest, fake DB connection, Flask test client
Documentation	Markdown source, python-docx, ReportLab, python-pptx

9. System architecture



System architecture

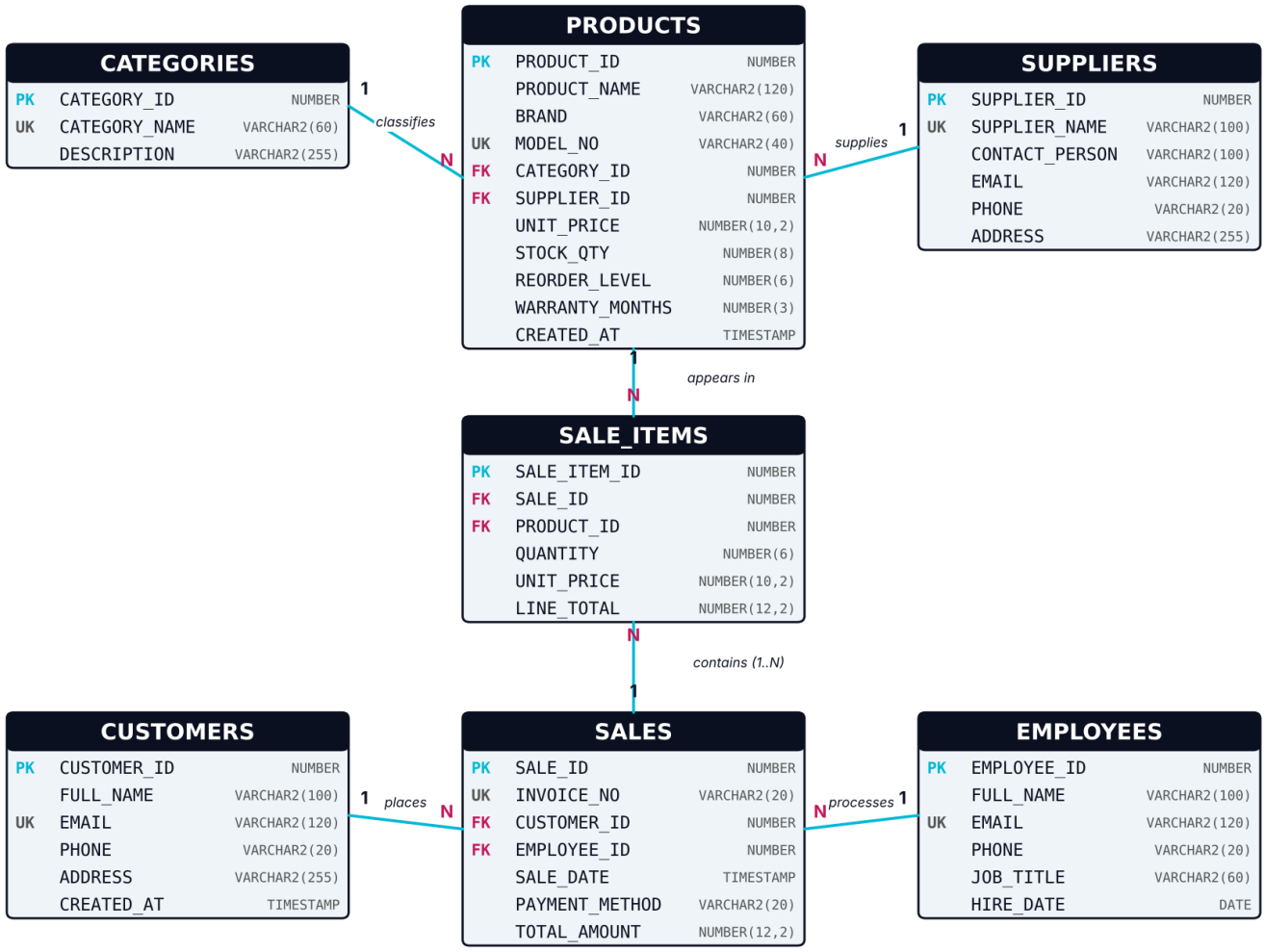
Routes call services only; services own all SQL; the database layer is the only module importing `oracledb`'s connect call.

10. Database design

Naming: upper-case Oracle identifiers; `PK_`, `FK_`, `UQ_`, `CK_`, `IX_` prefixes. Surrogate keys use `NUMBER GENERATED ALWAYS AS IDENTITY`. Money columns are `NUMBER(p,2)` (Python `Decimal`, never float). Timestamps use `TIMESTAMP`. Deletion behaviour: every foreign key is `NO ACTION` (restrict) — there is no cascade, so historical sales cannot be destroyed by deleting a customer, employee or product (ORA-02292,

reported to users as a friendly 409). Indexes: FK columns (`IX_PRODUCTS_CATEGORY`, `IX_PRODUCTS_SUPPLIER`, `IX_SALES_CUSTOMER`, `IX_SALES_EMPLOYEE`, `IX_SALE_ITEMS_PRODUCT`) and the report filter `IX_SALES_DATE`; PK/UNIQUE constraints supply their own indexes (including `UQ_SALE_ITEMS_LINE (SALE_ID, PRODUCT_ID)` which serves lookups by sale).

11. ER diagram



PK primary key FK foreign key (all NO ACTION / restrict, no cascade) UK unique key
Also UNIQUE (SALE_ID, PRODUCT_ID) on SALE_ITEMS. Drawn from database/schema.sql.

ER diagram generated from database/schema.sql

Mermaid source of the same diagram (renders on GitHub; omitted from the PDF/DOCX where the image above is used):

12. Table descriptions and relationships

Table	Purpose	Key columns
CATEGORIES	Product groups	PK `CATEGORY_ID`; UK `CATEGORY_NAME`
SUPPLIERS	Vendors	PK `SUPPLIER_ID`; UK `SUPPLIER_NAME`
PRODUCTS	Central product catalogue and live stock	PK `PRODUCT_ID`; UK `MODEL_NO`; FK `CATEGORY_ID`, `SUPPLIER_ID`
CUSTOMERS	Buyers	PK `CUSTOMER_ID`; UK `EMAIL`
EMPLOYEES	Staff who process sales	PK `EMPLOYEE_ID`; UK `EMAIL`
SALES	One row per invoice	PK `SALE_ID`; UK `INVOICE_NO`; FK `CUSTOMER_ID`, `EMPLOYEE_ID`
SALE_ITEMS	Lines of a sale with historical price/quantity	PK `SALE_ITEM_ID`; UK (`SALE_ID`,`PRODUCT_ID`); FK `SALE_ID`, `PRODUCT_ID`

Relational schema (PK underlined by convention, → marks a foreign key):

```
CATEGORIES (CATEGORY_ID, CATEGORY_NAME, DESCRIPTION)
SUPPLIERS  (SUPPLIER_ID, SUPPLIER_NAME, CONTACT_PERSON, EMAIL, PHONE, ADDRESS)
PRODUCTS   (PRODUCT_ID, PRODUCT_NAME, BRAND, MODEL_NO, CATEGORY_ID→CATEGORIES, SUPPLIER_ID→SUPPLIERS,
            UNIT_PRICE, STOCK_QTY, REORDER_LEVEL, WARRANTY_MONTHS, CREATED_AT)
CUSTOMERS  (CUSTOMER_ID, FULL_NAME, EMAIL, PHONE, ADDRESS, CREATED_AT)
EMPLOYEES  (EMPLOYEE_ID, FULL_NAME, EMAIL, PHONE, JOB_TITLE, HIRE_DATE)
SALES      (SALE_ID, INVOICE_NO, CUSTOMER_ID→CUSTOMERS, EMPLOYEE_ID→EMPLOYEES, SALE_DATE,
            PAYMENT_METHOD, TOTAL_AMOUNT)
SALE_ITEMS (SALE_ITEM_ID, SALE_ID→SALES, PRODUCT_ID→PRODUCTS, QUANTITY, UNIT_PRICE, LINE_TOTAL)
```

Relationships (all one-to-many): a category has many products; a supplier supplies many products; a customer has many sales; an employee processes many sales; a sale contains one or more items (≥1 enforced by the service); a product appears in many sale items. A product row is never duplicated by a sale — `SALE_ITEMS` references it and keeps the price and quantity at the time of sale.

13. Primary keys, foreign keys and integrity constraints

Table	Constraints (names as in `schema.sql`)
CATEGORIES	`PK_CATEGORIES`, `UQ_CATEGORIES_NAME`, `CATEGORY_NAME NOT NULL`
SUPPLIERS	`PK_SUPPLIERS`, `UQ_SUPPLIERS_NAME`, `SUPPLIER_NAME NOT NULL`
CUSTOMERS	`PK_CUSTOMERS`, `UQ_CUSTOMERS_EMAIL`, `FULL_NAME/EMAIL/PHONE NOT NULL`
EMPLOYEES	`PK_EMPLOYEES`, `UQ_EMPLOYEES_EMAIL`, `FULL_NAME/EMAIL/JOB_TITLE/HIRE_DATE NOT NULL`
PRODUCTS	`PK_PRODUCTS`, `UQ_PRODUCTS_MODEL`, `FK_PRODUCTS_CATEGORY`, `FK_PRODUCTS_SUPPLIER`, `CK_PRODUCTS_PRICE (UNIT_PRICE >= 0)`, `CK_PRODUCTS_STOCK (STOCK_QTY >= 0)`, `CK_PRODUCTS_REORDER (>= 0)`, `CK_PRODUCTS_WARRANTY (0..120)`
SALES	`PK_SALES`, `UQ_SALES_INVOICE`, `FK_SALES_CUSTOMER`, `FK_SALES_EMPLOYEE`, `CK_SALES_PAYMENT (CASH, CARD, UPI, BANK_TRANSFER)`, `CK_SALES_TOTAL (TOTAL_AMOUNT > 0)`
SALE_ITEMS	`PK_SALE_ITEMS`, `UQ_SALE_ITEMS_LINE (SALE_ID, PRODUCT_ID)`, `FK_SALE_ITEMS_SALE`, `FK_SALE_ITEMS_PRODUCT`, `CK_SALE_ITEMS_QTY (> 0)`, `CK_SALE_ITEMS_PRICE (>= 0)`, `CK_SALE_ITEMS_TOTAL (LINE_TOTAL = QUANTITY * UNIT_PRICE)`

E-mail and phone *format* validation is performed at the application level (`services/validation.py`) as specified. Sequence `SEQ_INVOICE_NO` (start 1000) generates invoice numbers `INV-001000`…; seed invoices use `INV-000001`… so they never collide.

14. Normalization analysis (1NF → 3NF)

1NF. Every column holds a single atomic value and no table has repeating groups: the "many products in one sale" requirement is modelled by rows in `SALE_ITEMS` rather than columns such as `PRODUCT1, PRODUCT2`. Each table has a primary key. (An address is kept as one atomic text attribute because the application never queries its parts; splitting it would be a refinement, not a 1NF requirement here.)

2NF. All tables use a single-column surrogate primary key, so partial dependencies on part of a key cannot occur. For `SALE_ITEMS` the natural candidate key is the composite (`SALE_ID`, `PRODUCT_ID`): `QUANTITY` and `UNIT_PRICE` depend on *both* (how many of which product on which invoice). Product attributes such as name or category are deliberately **not** copied into `SALE_ITEMS`; they would depend on `PRODUCT_ID` alone (a partial dependency) and are kept only in `PRODUCTS`.

3NF. No non-key attribute depends on another non-key attribute, with the examples:

Table	Functional dependencies	Why in 3NF
CATEGORIES	`CATEGORY_ID → CATEGORY_NAME, DESCRIPTION`; `CATEGORY_NAME → CATEGORY_ID`	Both determinants are candidate keys
SUPPLIERS	`SUPPLIER_ID → all`; `SUPPLIER_NAME → all`	Determinants are keys
PRODUCTS	`PRODUCT_ID → all`; `MODEL_NO → all`	Category and supplier *names* are not stored — only FKs — so no `PRODUCT_ID → CATEGORY_ID → CATEGORY_NAME` transitive dependency
CUSTOMERS	`CUSTOMER_ID → all`; `EMAIL → all`	Determinants are keys
EMPLOYEES	`EMPLOYEE_ID → all`; `EMAIL → all`	Determinants are keys
SALES	`SALE_ID → all`; `INVOICE_NO → all`	Customer/employee names are not stored (only FKs)

Table	Functional dependencies	Why in 3NF
SALE_ITEMS	SALE_ITEM_ID → all; (SALE_ID, PRODUCT_ID) → QUANTITY, UNIT_PRICE, LINE_TOTAL	See the documented exception below

Documented, deliberate redundancy (not hidden).

• SALE_ITEMS.LINE_TOTAL = QUANTITY × UNIT_PRICE is a derived column: {QUANTITY, UNIT_PRICE} → LINE_TOTAL has a non-key determinant, so strictly SALE_ITEMS is *not* in 3NF. It is stored for audit and fast reporting, and the anomaly risk is removed by CK_SALE_ITEMS_TOTAL.

• SALES.TOTAL_AMOUNT equals the sum of its lines' LINE_TOTAL. It is computed once by the service from database prices; V_SALES_SUMMARY.TOTAL_MATCHES exposes any mismatch. A CHECK cannot span tables, so this one is enforced by the service, not by a constraint.

• SALE_ITEMS.UNIT_PRICE repeats the *name* of PRODUCTS.UNIT_PRICE but not its meaning: it is the price at the moment of sale (history), whereas the product price is the current price.

15. Implementation details

• database/schema.sql — re-runnable DDL via an anonymous block that tolerates only ORA-00955.

• database/seed.sql — generated from fixed data; MERGE by natural keys; totals computed, not hand-typed.

• database/queries.sql — 16 numbered queries; database/advanced_features.sql — four objects plus self-rolling-back verification blocks; database/setup_user.sql — one-time admin script (password typed at a hidden prompt).

• Sample data (seed.sql, fictional; 6 categories, 4 suppliers, 14 products, 8 customers, 4 employees, 8 sales, 15 sale items). Examples:

Model	Product	Price (■)	Stock
ARC-RF260	Arctiva 260L Double Door Refrigerator	28,999.00	12
WW-FL70	WashWell 7kg Front Load Washing Machine	31,999.00	6
CB-AC10	CoolBreeze 1.0 Ton Window AC	26,499.00	2 (below reorder level 3)
VV-TV55	VistaView 55in 4K Smart TV	49,999.00	4 (at reorder level 4)
KP-MG75	KitchenPro Mixer Grinder 750W	3,299.00	25

Sale INV-000001 (Anita Rao, card): 1 × ARC-RF260 + 2 × KP-MG75 = ■35,597.00. The eight seeded sales total ■339,582.00 (hand-checked against the seed values; not yet confirmed by a live query). Two products (VV-TV32, AU-EK18) are intentionally never sold, and one customer has no sales, to give the NOT EXISTS and LEFT JOIN queries meaningful output.

• Python: config.py; services/*; app.py (25 routes); Jinja templates and static assets.

16. Flask and Oracle connectivity

config.load_db_settings() reads DB_HOST, DB_PORT, DB_SERVICE_NAME, DB_USER, DB_PASSWORD from the environment (python-dotenv loads .env at app start). Values still in <placeholder> form count as missing; errors list variable *names* only; SYS/SYSTEM/SYSDBA/SYSOPER are refused. services.database.get_connection() opens a Thin-mode connection per unit of work (oracledb.connect(user, password, dsn="host:port/XEPDB1")) and connection() always closes it (rolling back on error). oracledb.defaults.fetch_decimals = True returns money as Decimal. Diagnostics: python -m services.database and GET /health — both report success only after SELECT USER … FROM DUAL returned. Unavailable Oracle yields HTTP 503 with remediation text and no invented figures.

17. CRUD operations

Products, categories, suppliers and customers each have list (search/filter/pagination), create, update (edit form pre-filled via ?edit=<id>) and delete (POST, confirmation dialog, CSRF). Each service validates input, then runs a parameterized statement in its own short transaction. Constraint errors are translated: ORA-00001 → 409 "already exists", ORA-02291 → 400 invalid reference, ORA-02292 → 409 "referenced by other records",

ORA-02290 → 400 rule violation, ORA-30006 → 409 "inventory busy". Restock calls SP_RESTOCK_PRODUCT.

18. Transaction management and inventory consistency

services/sales.create_sale (single connection, no autocommit):

• Validate ids, payment method and items (non-empty, quantity 1–1000, ≤50 lines; duplicate products merged). No SQL runs if validation fails.

• Verify the customer and employee exist.

• For each product **in ascending id order** run SELECT … FOR UPDATE WAIT 10 (consistent lock order → no deadlock between sales; bounded wait → ORA-30006 instead of hanging).

• After *all* locks are held, re-check stock; any shortage → InsufficientStockError listing the products.

• Compute each line and the total as Decimal from the **locked database rows** (client prices/totals are never accepted — create_sale has no price/total parameters).

• INSERT the sale (invoice from SEQ_INVOICE_NO, RETURNING SALE_ID), INSERT each item, UPDATE PRODUCTS SET STOCK_QTY = STOCK_QTY - :qty — the **only** place stock is deducted (the trigger deliberately does not touch inventory).

• One COMMIT. Any exception → rollback (the connection() context manager and the connection close both enforce it).

Oracle's default READ COMMITTED isolation plus row locks prevents overselling: a concurrent buyer blocks on the lock, then re-reads the committed lower stock. CK_PRODUCTS_STOCK is the final safety net. Unit tests prove ordering, single commit, rollback and parameterization against a fake connection; the live concurrency proof is integration test INT-16 "concurrent sales cannot oversell" (**PASSED** on Oracle 21c XE).

19. SQL joins, subqueries and aggregations

#	Query	Concept
1	In-stock products over ■20,000	SELECT, WHERE, ORDER BY
2	Products in categories matching a pattern	IN subquery
3	Products with category	INNER JOIN
4	Products with supplier	JOIN
5	Customers with their sales (incl. none)	LEFT JOIN
6	Employee sales performance	LEFT JOIN + COUNT/SUM
7	Full invoice lines	5-table JOIN
8	Products per category	GROUP BY + COUNT
9	Monthly revenue	SUM/COUNT/AVG by date expression
10	Categories with revenue > 50,000	HAVING
11	Products above average price	Scalar subquery
12	Products above their category average	Correlated subquery
13	Products never sold	NOT EXISTS
14	Top 5 products by units	Aggregation + FETCH FIRST
15	Customer spend ranking	RANK() analytic
16	Inventory value and reorder status	CASE KPI

Example queries (verbatim from database/queries.sql)

```
-- Query 7: Full invoice lines (five-table JOIN)
-- Concept: SALES - CUSTOMERS - EMPLOYEES - SALE_ITEMS - PRODUCTS.
SELECT s.INVOICE_NO, TO_CHAR(s.SALE_DATE, 'YYYY-MM-DD') AS SALE_DAY,
       c.FULL_NAME AS CUSTOMER, e.FULL_NAME AS EMPLOYEE,
       p.PRODUCT_NAME, si.QUANTITY, si.UNIT_PRICE, si.LINE_TOTAL
  FROM SALES s
  JOIN CUSTOMERS  c  ON c.CUSTOMER_ID = s.CUSTOMER_ID
  JOIN EMPLOYEES  e  ON e.EMPLOYEE_ID = s.EMPLOYEE_ID
  JOIN SALE_ITEMS si ON si.SALE_ID    = s.SALE_ID
  JOIN PRODUCTS   p  ON p.PRODUCT_ID  = si.PRODUCT_ID
 ORDER BY s.SALE_DATE, s.INVOICE_NO, p.PRODUCT_NAME;
```

```
-- Query 10: Categories with revenue above 50,000
-- Concept: HAVING filters groups after aggregation.
SELECT c.CATEGORY_NAME, SUM(si.LINE_TOTAL) AS CATEGORY_REVENUE
  FROM SALE_ITEMS si
  JOIN PRODUCTS   p ON p.PRODUCT_ID  = si.PRODUCT_ID
  JOIN CATEGORIES c ON c.CATEGORY_ID = p.CATEGORY_ID
 GROUP BY c.CATEGORY_ID, c.CATEGORY_NAME
HAVING SUM(si.LINE_TOTAL) > 50000
 ORDER BY CATEGORY_REVENUE DESC;

-- Query 12: Products priced above the average of their OWN category
-- Concept: correlated subquery (inner query references the outer row).
SELECT p.PRODUCT_NAME, p.CATEGORY_ID, p.UNIT_PRICE
  FROM PRODUCTS p
 WHERE p.UNIT_PRICE > (SELECT AVG(p2.UNIT_PRICE)
                         FROM PRODUCTS p2
                        WHERE p2.CATEGORY_ID = p.CATEGORY_ID)
 ORDER BY p.CATEGORY_ID, p.UNIT_PRICE DESC;

-- Query 13: Products that have never been sold
-- Concept: NOT EXISTS anti-join (dead stock report).
SELECT p.PRODUCT_ID, p.PRODUCT_NAME, p.STOCK_QTY
  FROM PRODUCTS p
 WHERE NOT EXISTS (SELECT 1 FROM SALE_ITEMS si WHERE si.PRODUCT_ID = p.PRODUCT_ID)
 ORDER BY p.PRODUCT_NAME;

-- Query 15: Customer spending ranking
-- Concept: LEFT JOIN, aggregation and the RANK() analytic function.
SELECT c.FULL_NAME, COUNT(s.SALE_ID) AS PURCHASES,
       NVL(SUM(s.TOTAL_AMOUNT), 0) AS TOTAL_SPENT,
       RANK() OVER (ORDER BY NVL(SUM(s.TOTAL_AMOUNT), 0) DESC) AS SPEND_RANK
  FROM CUSTOMERS c
  LEFT JOIN SALES s ON s.CUSTOMER_ID = c.CUSTOMER_ID
 GROUP BY c.CUSTOMER_ID, c.FULL_NAME
 ORDER BY SPEND_RANK, c.FULL_NAME;
```

Status: all 16 statements parse with `sqlglot` (Oracle dialect) — a static syntax check; in addition all 16 queries were **executed live on Oracle 21c XE** (2026-10-22) without error. With the seed data Q13 should return `VV-TV32` and `AU-EK18`, Q10 should list Televisions (■139,996), Air Conditioners (■63,498) and Washing Machines (■50,998), Q16 flags `CB-AC10` ($2 \le 3$) and `VV-TV55` ($4 \le 4$) as REORDER NOW (expected values derived by hand from the seed). The live run matched Q13, Q16 and the Q10 Televisions and Washing Machines figures; Q10 Air Conditioners showed ■100,497 instead of ■63,498 because the demo database also holds one extra sale (INV-001000, CoolBreeze 1.5 Ton Split AC ■36,999) recorded through the web application.

20. View, procedure, function and trigger

Object	Purpose	Errors / behaviour
View `V_SALES_SUMMARY`	One row per sale with customer, employee, line/unit counts and `TOTAL_MATCHES`	Sales without items still appear (LEFT JOIN); used by sales history and dashboard
Procedure `SP_RESTOCK_PRODUCT(p_product_id, p_quantity)`	Add stock under a row lock; does not commit (caller owns the transaction)	ORA-20001 invalid quantity, ORA-20002 unknown product; used by the Restock button
Function `FN_CUSTOMER_TOTAL_SPEND(p_customer_id)`	Lifetime spend of a customer, 0 if none	ORA-20010 unknown customer; callable from SQL
Trigger `TRG_SALES_PROTECT_HISTORY`	Integrity: blocks DELETE of a sale and UPDATE of customer, employee, invoice, date, total, payment	ORA-20030 / ORA-20031; no inventory logic; documented test-cleanup bypass via `CLIENT_INFO`

```
CREATE OR REPLACE VIEW V_SALES_SUMMARY AS
SELECT s.SALE_ID, s.INVOICE_NO, s.SALE_DATE, s.PAYMENT_METHOD,
       s.CUSTOMER_ID, c.FULL_NAME AS CUSTOMER_NAME,
       s.EMPLOYEE_ID, e.FULL_NAME AS EMPLOYEE_NAME,
       COUNT(si.SALE_ITEM_ID)        AS LINE_COUNT,
       NVL(SUM(si.QUANTITY), 0)      AS UNIT_COUNT,
       s.TOTAL_AMOUNT,
       CASE WHEN s.TOTAL_AMOUNT = NVL(SUM(si.LINE_TOTAL), 0) THEN 'Y' ELSE 'N' END AS TOTAL_MATCHES
  FROM SALES s
  JOIN CUSTOMERS c ON c.CUSTOMER_ID = s.CUSTOMER_ID
  JOIN EMPLOYEES e ON e.EMPLOYEE_ID = s.EMPLOYEE_ID
  LEFT JOIN SALE_ITEMS si ON si.SALE_ID = s.SALE_ID
 GROUP BY s.SALE_ID, s.INVOICE_NO, s.SALE_DATE, s.PAYMENT_METHOD,
          s.CUSTOMER_ID, c.FULL_NAME, s.EMPLOYEE_ID, e.FULL_NAME, s.TOTAL_AMOUNT;
```

```
CREATE OR REPLACE PROCEDURE SP_RESTOCK_PRODUCT (
  p_product_id IN NUMBER,
  p_quantity   IN NUMBER
) AS
  v_stock PRODUCTS.STOCK_QTY%TYPE;
BEGIN
```

```
  IF p_quantity IS NULL OR p_quantity <= 0 OR p_quantity <> TRUNC(p_quantity) THEN
    RAISE_APPLICATION_ERROR(-20001, 'Restock quantity must be a positive whole number.');
  END IF;

  SELECT STOCK_QTY INTO v_stock
    FROM PRODUCTS
   WHERE PRODUCT_ID = p_product_id
     FOR UPDATE;

  UPDATE PRODUCTS SET STOCK_QTY = v_stock + p_quantity
   WHERE PRODUCT_ID = p_product_id;
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    RAISE_APPLICATION_ERROR(-20002, 'Product ' || p_product_id || ' does not exist.');
END SP_RESTOCK_PRODUCT;
```

```
CREATE OR REPLACE FUNCTION FN_CUSTOMER_TOTAL_SPEND (
  p_customer_id IN NUMBER
) RETURN NUMBER AS
  v_exists NUMBER;
  v_total  NUMBER;
BEGIN
  SELECT COUNT(*) INTO v_exists FROM CUSTOMERS WHERE CUSTOMER_ID = p_customer_id;
  IF v_exists = 0 THEN
    RAISE_APPLICATION_ERROR(-20010, 'Customer ' || p_customer_id || ' does not exist.');
  END IF;
  SELECT NVL(SUM(TOTAL_AMOUNT), 0) INTO v_total
    FROM SALES WHERE CUSTOMER_ID = p_customer_id;
  RETURN v_total;
END FN_CUSTOMER_TOTAL_SPEND;
```

```
CREATE OR REPLACE TRIGGER TRG_SALES_PROTECT_HISTORY
BEFORE DELETE OR UPDATE OF CUSTOMER_ID, EMPLOYEE_ID, INVOICE_NO, SALE_DATE, TOTAL_AMOUNT, PAYMENT_METHOD
ON SALES
FOR EACH ROW
BEGIN
  IF SYS_CONTEXT('USERENV', 'CLIENT_INFO') = 'ITEST_CLEANUP' THEN
    RETURN;
  END IF;
  IF DELETING THEN
    RAISE_APPLICATION_ERROR(-20030, 'Sales records are historical and cannot be deleted.');
  ELSE
    RAISE_APPLICATION_ERROR(-20031, 'Posted sales cannot be modified.');
  END IF;
END TRG_SALES_PROTECT_HISTORY;
```

Each has example invocation and verification SQL in `advanced_features.sql` (the verification blocks roll back). Live: all four objects load as VALID and integration tests INT-03 and INT-08..11 **PASSED** on Oracle 21c XE.

21. CRUD and sales screens (screenshots)

The Flask routes render server-side forms for the CRUD operations and the sale transaction. Each form posts to a real route that calls a service module; there are no mock stubs in the application code.

How these screenshots were produced. No Oracle database was available. Figures 1–3 show the *real* Flask routes and Jinja templates rendered in headless Chromium with the service functions replaced by seed-equivalent sample data; they demonstrate the CRUD and sale screens, **not** live database results. Figure 4 is the unmodified application started without database configuration and shows how a connectivity/configuration failure is reported (HTTP 503, no invented figures).

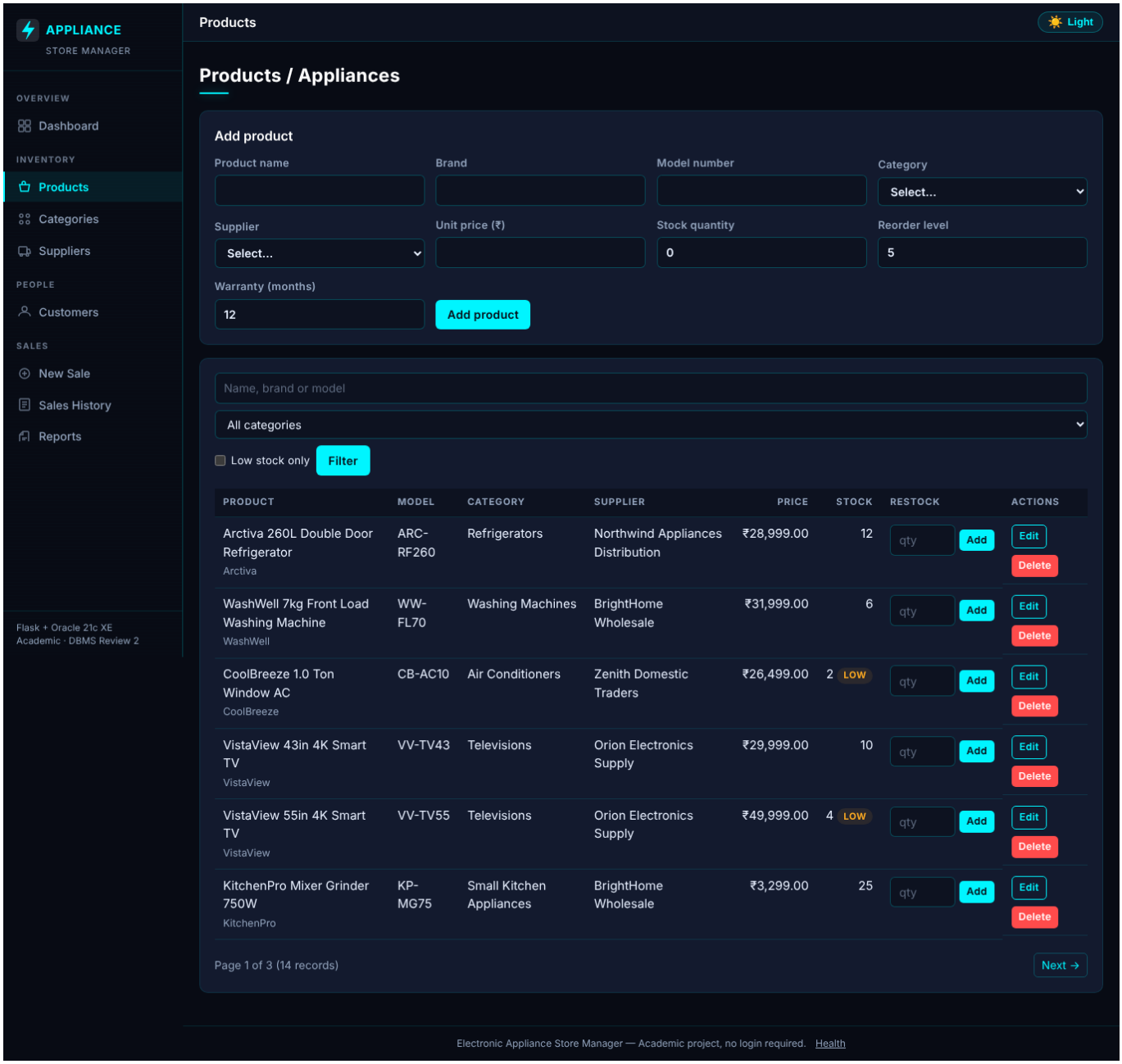


Figure 1 – Products: create form, search/filter, edit, delete and restock (sample data)

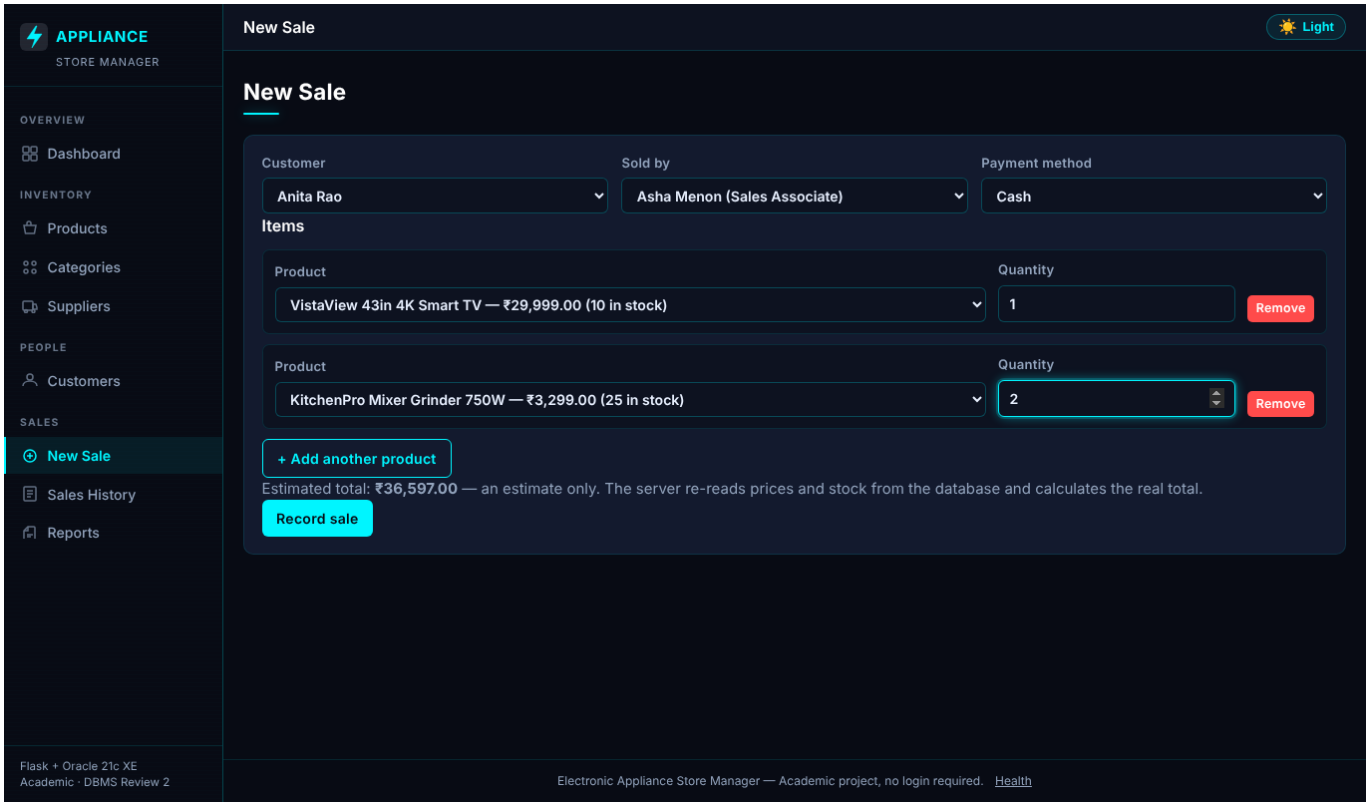


Figure 2 – New sale with two lines; the total shown is only an estimate, the server recalculates from database prices (sample data)

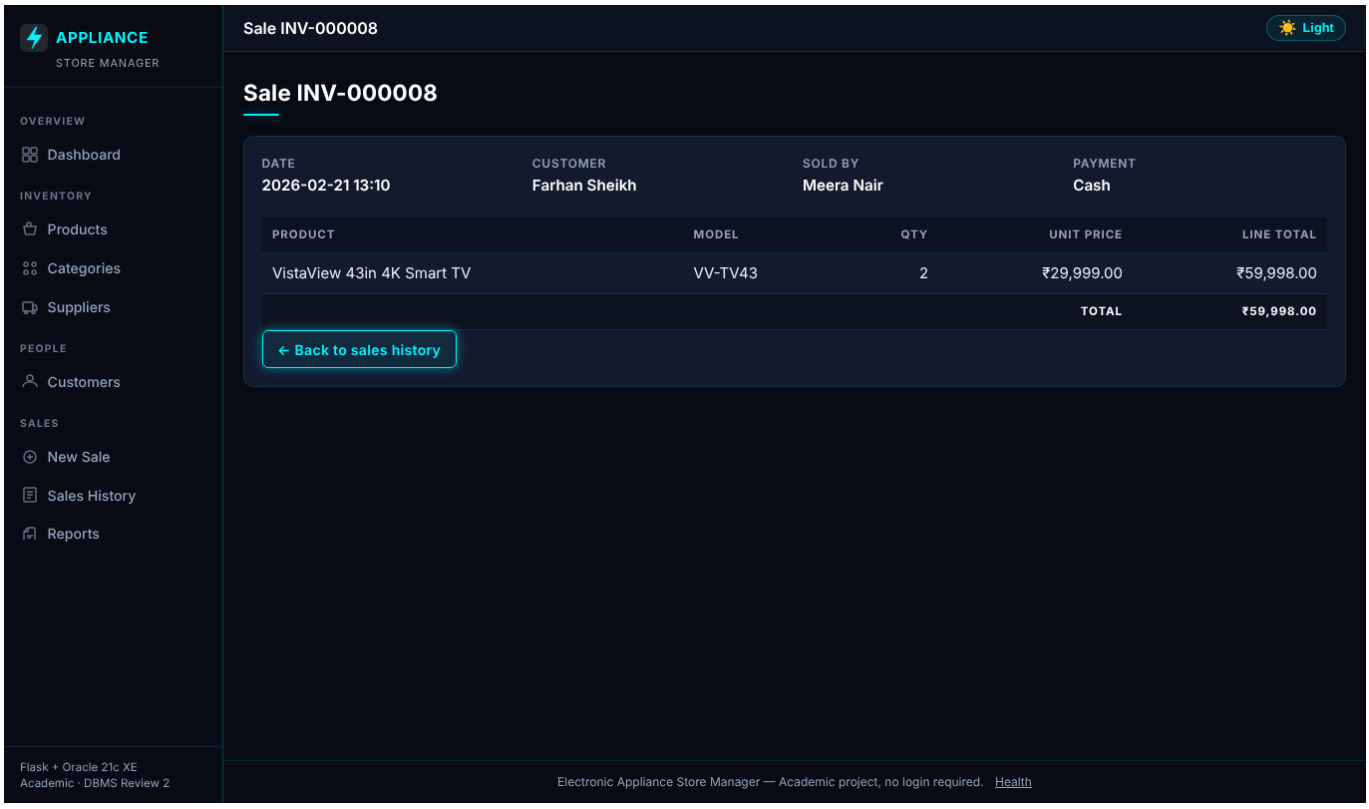


Figure 3 – Sale detail after a sale (sample data)

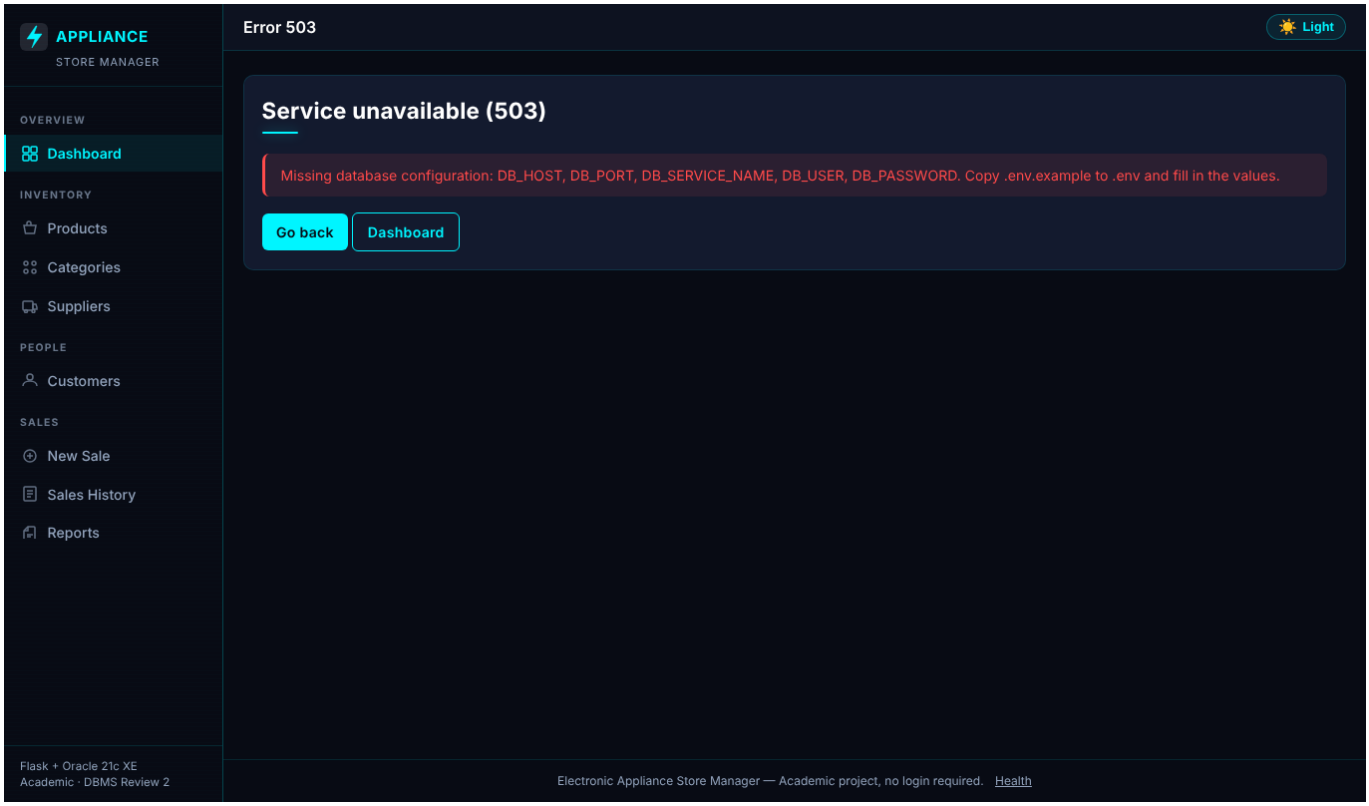


Figure 4 – Real application without a configured database: honest 503 page

22. Database access safeguards

• All SQL uses bind variables; user text never enters a statement (injection test SVC- in the matrix).

• The application connects only as the least-privilege user `appliance_app` (six CREATE privileges, no DBA); configuration rejects SYS/SYSTEM; administrator setup is the separate manual script `database/setup_user.sql`.

• Database errors are translated to safe messages (ORA codes, SQL and credentials go only to the server log).

• CHECK, UNIQUE and FK constraints reject negative price/stock/quantity and protect history; the trigger blocks edits to posted sales.

• Login/roles are not part of Review 2 (see §26).

23. Testing methodology

pytest suites: `test_config` (settings), `test_services` (validation, CRUD and transaction logic against a fake connection that records SQL, parameters, commits, rollbacks and closes), `test_routes` (Flask test client with stubbed services: CSRF, status codes, escaping, headers, health), `test_oracle_integration` (live Oracle, separate test schema, tagged rows, explicit opt-in). Status vocabulary — **PASSED**: executed and passed · **FAILED**: executed and failed · **NOT RUN**: intentionally skipped · **BLOCKED**: required dependency unavailable. Fakes prove application logic, not Oracle semantics (locking, constraint enforcement, PL/SQL), which is why the integration suite exists. Mutation sanity checks: removing the lock-order sort, and adding a mid-sale commit, each made tests fail (then reverted).

24. Verified testing matrix

Generated from the actual pytest result file (`docs/test_results.xml`) by `docs/build_reports.py`.

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
CFG-01	Valid DB_* variables produce settings and the XEPDB1 DSN	host:port/service DSN, no error	Executed; assertions held	**PASSED**	Pure logic
CFG-02	Missing DB_PASSWORD and DB_HOST are reported by NAME only	ConfigError lists names, leaks no value	Executed; assertions held	**PASSED**	Pure logic

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
CFG-03	Values copied unchanged from .env.example (<set-locally>)	treated as missing configuration	Executed; assertions held	**PASSED**	Pure logic
CFG-04	DB_USER set to an administrator account (3 parameter sets)	ConfigError (app must use appliance_app only)	Executed; assertions held	**PASSED**	Pure logic
CFG-05	Non-numeric or out-of-range DB_PORT (3 parameter sets)	ConfigError	Executed; assertions held	**PASSED**	Pure logic
CFG-06	repr() of the settings object	does not contain the password	Executed; assertions held	**PASSED**	Pure logic
CFG-07	FLASK_SECRET_KEY missing or placeholder	ConfigError with generation hint	Executed; assertions held	**PASSED**	Pure logic
CFG-08	APP_ENV=production	secure cookies on, debug off even if FLASK_DEBUG=1	Executed; assertions held	**PASSED**	Pure logic
SVC-01	Valid monetary strings (3 parameter sets)	accepted and quantised to 2 decimals	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-02	Negative, non-numeric, 3-decimal, NaN, too-large prices (6 parameter sets)	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-03	Malformed email / phone numbers	ValidationError; valid ones accepted	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-04	Zero, negative, non-integer, None, absurd IDs (6 parameter sets)	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-05	Duplicate products merged, lock order ascending	[(1, 3), (5, 2)]	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-06	Blank form rows ignored; only blanks -> empty sale rejected	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-07	Zero, negative, non-numeric, fractional or huge quantity (5 parameter sets)	ValidationError	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-08	Two-item sale	one commit, DB-priced total 35597.00 (Decimal), stock deducted per item, connection closed	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-09	Items supplied as 3,1,2	FOR UPDATE issued for 1,2,3 (consistent order -> no deadlock), all before any INSERT	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-10	Request 3 of a product with 2 in stock	InsufficientStockError, nothing inserted or updated, no commit, rolled back	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-11	Product id 999 does not exist	ValidationError, no writes, no commit	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-12	Customer id missing in DB	ValidationError before any product is locked	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-13	Stock UPDATE fails (CHECK violation) after sale+item inserted	ValidationError, rollback, NO commit	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-14	Bad payment method / empty items / bad ids (4 parameter sets)	ValidationError and zero SQL executed	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-15	Every statement uses bind variables; create_sale has no price/total input	no literals from user data in SQL	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-16	advanced_features.sql trigger body	contains no UPDATE of PRODUCTS (single deduction point)	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
SVC-17	SQL-injection text in a product name	stays in bind params, never in the SQL string	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-18	Negative price or stock in the form	ValidationError before any SQL	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-19	UPDATE affecting 0 rows	NotFoundError (404)	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-20	ORA-02292 on delete	ConflictError with friendly text, no ORA code/constraint name leaked	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-21	ORA-00001 on insert	ConflictError (409) with generic message	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-22	Listener down (ORA-12541)	DatabaseUnavailableError (503) without host/user/password details	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-23	Restock 10 units	SP_RESTOCK_PRODUCT called with (id, qty), exactly one commit	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-24	Procedure raises ORA-20002	NotFoundError, rolled back	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-25	oracledb.connect raising ORA-01017 (bad login)	DatabaseUnavailableError, no credentials in message	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
SVC-26	No DB configured	health_check returns (False, message) - never claims success	Executed; assertions held	**PASSED**	Fake DB connection - proves SQL/transaction logic, not Oracle behaviour
RTE-01	Dashboard with stubbed DB figures	200 and the figures are rendered	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-02	Oracle unavailable	HTTP 503 error page, no statistics shown	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-03	Category name containing <script>	rendered escaped, never as live markup	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-04	POST with no CSRF token	400 and the service is never called	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-05	POST with a forged CSRF token	400	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-06	Valid CSRF token + valid form	service called, 303 redirect (post/redirect/get)	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-07	X-CSRF-Token header instead of form field	accepted	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-08	GET on delete / update / restock URLs	405 Method Not Allowed	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-09	Service raises ValidationError	HTTP 400 with the message	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-10	Delete of a referenced row	HTTP 409 with friendly message	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-11	Sale rejected for insufficient stock	HTTP 409 with product detail	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-12	Forged price/total/employee fields in the sale POST	price and total ignored; employee id comes from the login session	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-13	Sale form with no product lines	HTTP 400, create_sale never reached	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-14	Unknown URL -> 404; non-numeric sale id -> 404	Unknown URL -> 404; non-numeric sale id -> 404	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-15	Unexpected exception text with a secret	HTTP 500, page shows a generic message only	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-16	Any response	CSP, nosniff and frame protection headers set	Executed; assertions held	**PASSED**	Services stubbed; Flask test client

ID	Test description	Expected outcome	Actual outcome	Status	Remarks
RTE-17	No Oracle configured	/health returns 503 and database=unavailable, never 'ok'	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-18	health_check reporting success	/health 200 with no secrets in the body	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
RTE-19	Products, customers, sales, sale detail, new sale, reports pages	all render 200 with stubbed data	Executed; assertions held	**PASSED**	Services stubbed; Flask test client
INT-01	Connect to XEPDB1 as the test application user	connected, user is not SYS/SYSTEM	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-02	USER_TABLES	all 7 required tables exist	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-03	USER_OBJECTS	view, procedure, function, trigger all VALID	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-04	Negative price/stock, duplicate model, bad category FK	ORA-02290 / ORA-00001 / ORA-02291	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-05	SALE_ITEMS with LINE_TOTAL != QUANTITY*UNIT_PRICE	ORA-02290	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-06	Delete category/customer/product that history references	ConflictError (ORA-02292 translated)	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-07	Create, read, update, delete a category and customer through the services	each step reflected in DB	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-08	SP_RESTOCK_PRODUCT	+5 stock on success; ORA-20001 for 0, ORA-20002 unknown product	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-09	FN_CUSTOMER_TOTAL_SPEND	0 before sales, equals sale total after, ORA-20010 for unknown customer	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-10	TRG_SALES_PROTECT_HISTORY	DELETE -> ORA-20030, UPDATE total -> ORA-20031	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-11	V_SALES_SUMMARY	shows the new sale with correct line/unit counts and TOTAL_MATCHES = 'Y'	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-12	Two-item sale	totals from DB prices (2*100.00+3*49.50=348.50), stock 10->8 and 5->2, items stored	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-13	Quantity above stock	InsufficientStockError; no sale row, stock unchanged	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-14	Valid line + nonexistent product	ValidationError and the valid line's stock is untouched	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-15	Another session holds FOR UPDATE on a product	sale waits then fails with ConflictError (ORA-30006), stock intact	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)
INT-16	8 threads each buy 1 unit of a product with 5 in stock	exactly 5 succeed, 3 InsufficientStockError, final stock 0	Executed; assertions held	**PASSED**	Live Oracle 21c XE, separate test schema (student run 2026-10-22)

Totals by pytest case: PASSED 92

Other checks executed (sandbox: Linux, Python 3.13, no Oracle)

Check	Command	Result
Python compilation	`python -m compileall -q app.py config.py services tests`	PASSED

Check	Command	Result
JavaScript syntax	`node --check static/js/app.js`	PASSED
Query count	`grep -c "^-- Query [0-9]*:" database/queries.sql`	16 — PASSED
SQL static parse	`sqlglot` (Oracle dialect) over 16 queries and 7 seed MERGE statements	PASSED (syntax only; PL/SQL scripts not parsed — NOT RUN)
Flask smoke	`python app.py` then `GET /health`, `GET /products`, `POST /categories` without token	503, 503, 400 as designed — PASSED
Connection diagnostic	`python -m services.database` with an unreachable listener	exit 1, sanitized message, no secrets — PASSED
Secret scan	recursive grep for credential-like assignments	only obviously fake test dummies; PASSED
Live Oracle: schema, seed, queries, PL/SQL, integration tests	README §7, §9	**PASSED** — run by the student on Oracle 21c XE, 2026-10-22: 16/16 integration tests, 16/16 queries (no `ORA-` errors)

25. Results and discussion

Everything that can run without Oracle ran and passed: configuration rules, validation, service logic including the full sale transaction path (lock order, stock re-check, single commit, rollback on failure), error translation (no leakage of credentials or ORA internals), route behaviour (400/404/409/503 statuses, honest health reporting, forms reaching the services). The design choice of one deduction point (service) and sorted row locks gives deterministic concurrency behaviour that is straightforward to test. The live Oracle run then confirmed the parts fakes cannot prove: acceptance of the DDL/PL/SQL scripts, constraint and trigger behaviour, the query results quoted in §19, and the row-lock and concurrency tests (INT-15, INT-16).

26. Known limitations

• No authentication/authorization or per-user audit trail.

• Screenshots use sample data behind the real templates (no live Oracle); only Figure 4 shows unmodified runtime behaviour.

• The live results come from one run on one machine (Oracle 21c XE, single user); no load or multi-machine testing was done.

• `SALES.TOTAL_AMOUNT` ↔ line totals consistency is service-enforced, not a constraint; `LINE_TOTAL` is a documented derived column.

• Trigger contains a test-cleanup bypass that should be removed for production-style use.

• No returns, discounts, taxes, multi-currency; employees have no CRUD screen.

• `LIKE` searches treat %/_ as wildcards; per-request connections (no pool) are adequate for a demo, not for load.

• The seed is first-run oriented: it never overwrites changed rows, so edits to seeded products persist.

27. Installation and execution summary

See README §4–§9. In brief (PowerShell): create `appliance_app` once as admin with `database\setup_user.sql`; `py -3.11 -m venv .venv`; activate; `pip install -r requirements.txt`; copy `.env.example` to `.env` and fill in the password and a generated secret key; run `schema.sql`, `advanced_features.sql`, `seed.sql` with `sqlplus appliance_app@localhost:1521/XEPDB1 @database\<file>.sql`; `python -m services.database`; `python -m pytest tests -v`; `python app.py`; open `http://127.0.0.1:5000`.

Demonstration (10 minutes, details in `docs/demo_guide.md`): (1) `python -m services.database` and open the app; (2) show `schema.sql` constraints and the failing inserts; (3) verify seed counts; (4) run queries Q7, Q10, Q12, Q13; (5) show the four PL/SQL objects; (6) `/health` and CRUD; (7) record a two-item sale and watch stock fall, then try an oversized sale (HTTP 409); (8) show the test matrix (76 offline and 16 live cases PASSED).

28. Faculty evaluation cross-reference

#	Criterion	Where implemented	How to demonstrate	Verification status
1	Database and table creation	`database/schema.sql`, `setup_user.sql`, README §5–§7	Run scripts; `SELECT table_name FROM user_tables`	Live: scripts ran on Oracle 21c XE; 7 tables confirmed by INT-02 — **PASSED**
2	Keys and constraints	PK/FK/UQ/NOT NULL/CHECK in `schema.sql`; §13	`user_constraints` query; demo_guide §2 failing inserts	Live enforcement confirmed by INT-04/05/06 — **PASSED**
3	Meaningful sample data	`database/seed.sql` (7 tables, 8 multi-item sales)	demo_guide §3	Live load ran on Oracle 21c XE; rows visible in query output — **PASSED**
4	Joins, subqueries, aggregations	`database/queries.sql` — exactly 16	Run file; demo_guide §4	Count checked; all 16 executed live without error — **PASSED**
5	Advanced database features	`database/advanced_features.sql` (view, procedure, function, trigger)	Verification blocks; demo_guide §5	Live PASSED (integration tests exist)
6	Backend–database connectivity	`config.py`, `services/database.py`, `/health`, `python -m services.database`	demo_guide §6	Config/error handling PASSED (unit); real connection INT-01 and the running app — **PASSED**
7	CRUD and transactions	`services/*.py`, `app.py`, `templates/*`, `services/sales.py`	UI workflow; demo_guide §7–§8	Logic PASSED with fakes; Oracle transactions and locking INT-07, INT-12..16 — **PASSED**

Live verification covers the database, queries, integration tests and CRUD/sales transactions as listed above. The browser walkthrough of the criteria table was not recorded as a formal test.

29. Conclusion

The project delivers a normalized seven-table Oracle design, deterministic data, sixteen documented queries, four PL/SQL-side objects, a layered Flask application with an atomic, lock-ordered sales transaction, and a status-honest test suite. All checks that can run without Oracle pass; the live Oracle checks were then run by the student on Windows against Oracle 21c XE (16/16 integration tests and 16/16 queries passed).

30. References

• Oracle Database 21c Documentation: SQL Language Reference; PL/SQL Language Reference; Database Concepts (data concurrency and consistency).

• python-oracledb documentation, <https://python-oracledb.readthedocs.io/>.

• Flask documentation, <https://flask.palletsprojects.com/>.

• OWASP Cheat Sheet Series: SQL Injection Prevention; Cross-Site Request Forgery Prevention.

• Elmasri, R. and Navathe, S., *Fundamentals of Database Systems*, Pearson — normalization chapters.